

The shaping of I.T. trajectories: evidence from the U.K. public sector

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Abstract

This paper reports upon I.T. initiatives in the U.K. public sector where choices were made, decisions taken, major investment authorised, and new I.T. introduced. The research investigation followed practice over several years in three Police Forces as new information technologies were introduced. We focus on these I.T. trajectories, and in particular the actions, interactions and events within and across groups of people. We argue that practices surrounding I.T. choice are socially shaped. Indeed, the choices and subsequent acquisition can be viewed as part of a process of ongoing configuration. We argue that Social Worlds Theory has much to offer in understanding what happens as I.T. trajectories evolve in organisations. The empirical part of the paper reports upon the findings from a longitudinal, cross-case analysis within the three Police Forces – institutions that are often characterised as bureaucratic in form, and culture. In each case, the I.T. trajectories were followed utilising a variety of qualitative techniques. Drawing upon social worlds theory, and boundary objects, we consider the influence of certain ‘objects’ upon I.T. outcomes, and the way in which other boundary objects become a focus for the construction of apparent facades of formality.

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Introduction

Introducing large-scale I.T. initiatives into any organisation is rarely straightforward. In the U.K. public sector there is increasing scrutiny, and public reporting of I.T. projects, often unfavourably, with calls for improved planning of these major I.T. investments (Arnott, 2003). These projects often utilise configurations of information and communication technologies, with the outcomes reflecting socio-technical interactions, as well as the culturally and historically situated nature of the usage (Avgerou & Madon, 2004). Earl (1996) has argued that many organisations successfully undertake large-scale investments in I.T. without necessarily having an explicit plan, while others have noted the existence of explicit plans (Jones, 1995). It has also been suggested that bureaucratic, or even autocratic contexts may be more likely to exhibit formal planning for I.T. (e.g. Peters *et al.*, 2002). In the U.K. public sector there are ongoing attempts (e.g. through The Gateway Process (Cabinet Office, 2000)) to encourage organisations to embrace a range of explicit, or formal techniques to assist the planning and evaluating of I.T. initiatives, as well as for monitoring the development/acquisition/appropriation/embedding process(es). It seems fair to suggest that encouraging such usage reflects an

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assumption that this use will lead to a successful outcome – a form of processual determinism it could be argued.

We have focused upon a public sector institutional setting that can be described as highly bureaucratic, with rigid hierarchies, and the potential for autocratic behaviour – the U.K. Police Service. We were interested in exploring the relationships between I.T. outcomes and socio-technical interaction; the configurational processes surrounding I.T. and organisation practice that have been described as being mutually constitutive (Mackenzie & Wajcman, 1999), that is, reflecting a process of mutual adaptation (Orlikowski, 1992). By this we infer that it is neither information, nor technology, nor organisational context, nor people alone that determine ‘an outcome’ – instead, we believe that it is through the mutual interaction between and within these areas that futures are shaped. We stress however, the observation that both historic and cultural facets contribute to these trajectories and must not be ignored. In this research, the oft-presented bi-polar distinction between planning and emergent behaviour influencing I.T. outcomes seemed inadequate as an explanatory device. It is the attention that we give as researchers to the interactions among people, and between people and I.T., as well as taking account of the particular historic and cultural situation that we believe to be essential to improving understanding of the trajectories of I.T. outcomes.

The study reported here investigated what happened over the course of several years as three public sector organisations each undertook a major I.T. initiative. We were interested in the relationship between organisational type and I.T. trajectories – not in the sense of trying to comprehend a deterministic path that leads to an outcome, but rather, in better understanding the range of actions and interactions that shape, in this instance I.T., outcomes (Strauss, 1993), as well as the historic and cultural settings of those actions and interactions. These trajectories are shaped within and across groups of people, being influenced by issues or matters both within and outwith the organisation. In order to help make sense of the detailed interactions as the trajectories evolved we sought a means of understanding both what happens in, and across those groups of people, as well as the influences upon these actions, interactions, and events. Social worlds theory (Clarke, 1991) proved valuable in this respect, and it is to this that we now turn.

Exploring socio-technical interaction with social worlds theory

Detailed studies investigating I.T. trajectories as practices and discourse reflecting sociotechnical interactions have become more numerous, although are still too few (Lamb & Sawyer, 2005). There has for some time been a view that the choices surrounding, and outcomes of, the utilisation of I.T. in organisations, are best appreciated by considering such I.T. initiatives as, ‘complex social objects constrained by their context, infrastructure and

history’ (Kling & Scacchi, 1982, p. 69). It has been argued that I.T. projects in bureaucratic contexts may be more likely to exhibit formal planning processes (e.g. Peters *et al.*, 2002). The possibility of a link between the nature of an institution and processes that may influence choices, and hence consequences, is interesting, particularly in the public sector. The public sector in the U.K. has seen growth in investment in I.T. accompanied by increasing attention given to many ‘failures’ in I.T. projects over recent decades – something that has been described as Repeated Failure Syndrome (RFS), or (Naughton, 2004). An article in the Economist on 16 October 2004 set out the ‘previous fiascos’ in public sector I.T. projects, bemoaning the failure to learn. Hence, improving our appreciation of I.T. trajectories and consequences in the public sector is important as a contribution to that process of learning.

One means of appreciating elements of uncertainty and unpredictability in I.T. projects is offered by symbolic interactionism (Blumer, 1967). From an organisational perspective, this is concerned with the way that meanings are constructed and reconstructed through workplace interactions between people in specific contexts. Three premises underpinning this view of interactionism (Gopal & Prasad, 2000) are instructive. Firstly, that actions are based upon meanings that people have derived from the situation. Secondly, that the meanings associated with social and non-social objects or symbols arise through interactions between people; and finally, that these meanings may be reconstructed as people continually interpret and reinterpret the developing situation. The challenge here is to make sense of both the interactions among people and to the range of social and non-social objects, all of which may influence the shaping of technological possibilities and outcomes.

The concept of trajectories can be subject to misinterpretation. It does not, in our usage, imply inevitability of outcome. Nor does it infer that the presence of specific factors will determine an outcome, or that a particular path of activity will continue in a foreseeable manner. Instead, the concept of trajectory draws our attention to the evolution of action *over time*, and to the actions and interactions contributing to that evolution (Strauss, 1993). Developing this idea in relation to I.T., projects may incorporate planning methods, use of analytic frameworks and software – all of which can be thought of as *trajectory schemes* (Garrety & Badham, 2000). These trajectory schemes ‘may shape the trajectory, but they do not *constitute its totality*... It is the inter-relationships between schemes and the contingencies that occur along the way that constitutes the politics of the project’ (*ibid.* p. 106, original emphasis). Thus, we have to look closely, and in detail, at the range of actions, interactions and schemes that contribute to the shaping of trajectories over time.

Social worlds theory (see Clarke, 1991) provided an opportunity for further insights, and this contributed considerably to our understanding of the evolving I.T.

trajectories. Clarke identifies the development of social worlds theory with Shibutani (1955; referenced in Clarke, *ibid.*) who focused upon the idea of commitment as a basis for social action. In our analysis, social worlds are groups with, 'shared commitments to the pursuit of a common task, who develop ideologies to define their work and who accumulate diverse resources needed to get the job done' (Gieryn, 1995, p. 412). Gieryn goes on to argue that social worlds have three properties in common, namely: the inevitability of intersection with other social worlds; the potential for division into what may be termed sub-worlds; and the practices of legitimisation where standards and boundaries are subject to construction and negotiation. Social worlds theory encourages us to inquire into how social worlds initiate, develop, and manage boundaries between worlds, and how worlds come to be legitimised (or otherwise). For our purposes, we are concerned with worlds relating to organisational settings, what Gerson (1983) terms *production worlds*.

In thinking about I.T. trajectories, the construction of facts, artefacts, as well as procedures can, it has been argued (Fujimura, 1992), be considered as collective processes that are temporally located. The activities of planning, evaluation, and coordination of actors usually become visible only when these activities are not done or are not done well (*ibid.*). One way of addressing this potential for focusing upon what could be termed the negative in seeking to isolate social worlds is to look at roles. This can help to distinguish between groups of people who may share certain sets of meanings, access to artefacts, and areas of activity (Gopal & Prasad, 2000). Roles reflect 'the socially defined expectations of behaviour for individuals in particular social positions', and provides, 'individuals with a complex set of identities, which become the source for individual interpretations of social situations' (*ibid.* p. 514). Examples of the ways in which roles may be differentiated include: hierarchy, status, expertise, interest, affiliation, and association. From this we can infer that roles may have a degree of temporal stability, such as where someone occupies a specific job. Equally, a role may be much less temporally stable, for example where there is a short-term requirement for expertise in an I.T. project.

Finally, the concept of boundary objects. Object refers to 'anything that can be indicated, anything that is pointed to or referred to' (Blumer, 1967, p. 10). Star & Griesemer (1989) define boundary objects as, 'objects which are both plastic enough to adapt to local needs and the constraints of the several parties employing them, yet robust enough to maintain a common identity across sites... . They have different meanings in different social worlds but their structure is common enough to more than one world to make them recognisable' (p. 393). Typically, these can be categorised into physical (artefacts, people), social (workgroupings, experts), and abstract objects (theory, power relations). Boundary object refers to those objects which are meaningful to several social worlds, and while the representation does

not change, the meanings derived may well differ across individuals and/or groups. We note that objectification must not become the focus of the exercise, as this would be to the detriment of understanding socio-technical actions and interactions.

In summary, drawing upon aspects of symbolic interactionism, and social worlds theory, offers us several means of comprehending the variety of actions and interactions that influence both the progress and outcome of I.T. trajectories. We now move on to consider the shaping of I.T. trajectories in three cases.

Empirical study – context and method

The Police Service is a part of the public sector in the U.K., and has been described as one of the last areas of the U.K. public sector to feel the touch of reform (Loveday, 1993). Increasingly, over recent times the Police have been required to become more accountable, representing a new public service focus (Jones & Smith, 1993). They have been subject to a series of Government initiatives to improve financial management, which have been at the heart of new public management (Barberis, 1998).

The majority (49) of the 52 Police Forces in the U.K. operate under a tripartite structure where operational control lies with the Chief Constable of each force. The Chief Constable is accountable both to a local Police Authority, as well as to a governmental Secretary of State (a senior political appointment in the U.K. Government). Funding (for most Forces) comes from both central and local government. The effectiveness, efficiency, and economy of operation of Police Forces is regularly assessed by Her Majesty's Inspectorate of Constabulary (with separate inspection bodies for England and Wales, and Scotland). Police Forces also subject to periodic scrutiny by the Audit Commission (e.g. Audit Commission, 1993), who monitor, and report upon the U.K. public sector. Police Force have been characterised as bureaucratic, even autocratic, but at the same time are seen to contain a considerable degree of discretion in operation (Reiner, 1994).

In terms of I.T., two areas of the wider environment are noteworthy: the first is the publication in 1994 of a National Information Systems Strategy for the Police in England and Wales (Home Office, 1994). This strategy document was supposed to inform the activities of Police Forces in relation to I.T. However, this strategy was subject to much debate and was subsequently abandoned, to be replaced with a new national strategy in 2002. This is planned for implementation by 2007 according to the Association of Chief Police Officers (ACPOS, 2002). The existence of the 1994 strategy was something that all of those spoken with during this study expressed awareness of. In Scotland, a Scottish Police Information Strategy (SPIS) commenced in 1993, with the first application becoming available in 2000. The second area of note is the existence of the Police I.T. organisation (PITO), a government agency set up in 1998, to advise Police Forces on I.T. use, and in particular to

oversee the National Information Systems Strategy. However, it has been criticised as having had little influence in encouraging Police Forces to adopt a national approach to I.T. initiatives (Collins, 2005).

In this respect, a few words on rank structure are required. The Police Service has a hierarchical role structure, with ranks. For the vast majority of Forces (including those reported here), the rank structure from most senior down are: Chief Constable, Deputy and Assistant Chief Constable, Chief Superintendent, Superintendent, Chief Inspector, Inspector, Sergeant, Constable. Many U.K. Police Forces have appointed non-police personnel to roles previously undertaken by ranks as senior as Assistant Chief Constable. The Chief Constable ranks (including Deputy and Assistant) can be regarded as forming the senior executive level of the organisation, although in some Forces officers from the Superintendent ranks may also play a part at this level (e.g. in smaller Police Forces). Similarly, senior non-police personnel may be a part of this executive level. The Chief/Inspector rank is normally regarded as a middle management position. Across the Police Service there has been a rise in the number of civilian staff into senior positions over the last 15 years, a move that has posed challenges within the structure and culture of these institutions. In relation to I.T., the importing of senior expertise has generally been regarded positively by police officers.

The empirical work reported here comprised three in-depth case studies, a recognised method for conducting research into socio-technical interactions (Darke *et al.*, 1998). The cases were conducted in a manner that reflected the principles for interpretive case studies set out by Klein & Myers (1999). The longitudinal, cross-case comparative analysis covers the period 1994–2000. Data collection involved a variety of qualitative techniques: 54 in-depth semi-structured interviews of 2 h duration on average that were tape recorded and transcribed; 18 semi-structured interviews of shorter duration (maximum 30 min) where notes were taken; numerous on- and off-site informal conversations, observation at meetings (during 1996); collection of documentation; collection of secondary materials produced outside of the direct areas of study. Many interviewees were spoken with several times during the course of the study. Data collection took place in each of the cases across the 6-year period. Latour (1987) suggests studying the ‘messiness’ of a situation; the sole investigator (first author) tried to do this by remaining sensitive to the emergence of practices and issues during the course of the research. Through the research process, including the analysis of data, an understanding of these other social worlds developed – that is, of the actions and interactions within and between worlds. It was not possible in this study to analyse precisely how the actors developed, directed and modified the various projects at the point when there were just ideas, or ideas under construction (Bowker & Star, 1999). However, the access to text documents and

the retrospective interviews that were undertaken contain reflections of the relationships and tensions between the (social) actors involved.

Our focus upon a single sector is deliberate, as this facilitated opportunities for comparing and contrasting practice (Pettigrew, 1985) across the social worlds, drawing attention to similarities or variations therein. The anonymity of participants and institutions has been preserved. Each of the cases is a summary of the large amount of data collected. However, the purpose of their presentation is to provide an appreciation of what happened in each, as a prelude to the discussion.

Case A

This Police Force was responsible for policing a city and a large area of surrounding countryside. Prior to 1993, the I.T. infrastructure at Force A was generally poor, having been ‘starved of development’ according to senior staff following the installation of a command and control system in 1983. Police officers in the Force had access, generally through one of a few divisional control rooms, to the few national I.T. systems that existed, but otherwise were dependent upon stand-alone PC-based systems for local applications such as databases.

At the end of May 1993 a new Deputy Chief Constable (the second most senior rank) arrived, and ordered the I.T. Department to prepare a specification to acquire a range of I.T. hardware and applications with a view to transforming the information system (I.S.) capability of the Force. The aim was to acquire off the shelf software applications as far as was possible in order to reduce delay in implementation, and cost, as far as possible.

At a meeting in mid-1993, the new Deputy Chief Constable referred to the need to have a strategy for I.T. in the Force. The development of an operational requirement for a new system was seen by those involved as just that, the development of an I.T. strategy. An operational requirement was drawn up by a group consisting of two mid-ranking officers and the computer manager. This group were in continual contact with the deputy Chief Constable, both through formal fortnightly meetings, but also through informal discussion on a sometimes daily basis. This informal communication took place more frequently between the Deputy Chief Constable, an Inspector and the computer manager; it was these three who were the main group responsible for developing the I.T. requirements. The operational requirement was a three-page document, and set out the current situation in the Force together with what was required in broad terms. As one person involved in this process commented,

we deliberately left it open so that when we are in the phase of the project where we are ready to look at what personnel system, we can then say ‘right, what is the best of breed that is round just now’, and not say ‘well four years ago we signed up to get that and really it has been overtaken ... so we tried to keep our options as flexible as we possibly can ... not without heartache because

the industry has found it difficult, they want to speak about deliverables and signing off parts of the contract'.

This operational requirement was couched in broad terms, and did not give detailed requirements beyond the required application titles. The rationale behind this was expressed as follows:

give the expert the problem and let him come up with the solution, don't give the expert the solution because you will get what you asked for.

A consequence of this decision was to prove interesting however; because in using the term 'best of breed', and not specifying precisely the application to be supplied, the supplier provided the police team with an area to debate when they eventually came to the date of supply – which is what happened. Hence, during 1995 and 1996, as applications came to be considered, the Police team demanded 'best of breed' systems at that point in time, and not what was best of breed when the contract was signed in February 1994. A supplier, who was an international I.T. vendor, was chosen following a competitive tendering process. All except one of the software applications to be adopted were already in existence, although each would require some customisation for the Force. The one application that was bespoke, a command and control system, was delayed when the Deputy Chief Constable decided after the contract had been signed to standardise the Force on Windows NT, rather than continue to use Unix – which had been the intended operating environment.

A project management board was set up in 1994 to oversee the project, and this met every 6 months for 3 years. The meetings were not documented in anyway, and were regarded as 'rubber stamping' exercises by those involved. The project formed the basis for I.T. development in the Force between 1994 and 1999. An acronym for the new suite of technology applications was devised, but a view expressed by many staff was of it as, '*the boss's I.T. project*'. Most of those closely involved were of the view that the project was the strategy, and vice versa. Among other senior Officers opinion varied; namely that there was not a written I.T. strategy as such, just a shopping list of applications, progress towards which had in their view been determined and driven by the deputy Chief Constable. However, as the Chief Inspector involved in developing the operational requirement commented, in 1996, when considering the I.T. initiative, 'it's all in our heads, we don't have time to sit down and write a strategy'. In 1998, an Inspector in that group observed that:

the strategy is really a hybrid. I think like everything else we had an idea that we would like to write down exactly where we were going ... we don't have as such a strategy document that we can hand out, we are working to the philosophy of strategy rather than to the word of it... So ... the documentation, there isn't a lot of it, we tend to be working by "hmmm that sounds a good idea let's try it and see", ... so I don't know how close that runs to

what we should be doing but, that's how we did it and it seemed to work.

During 1998 and 1999, the last few applications were implemented, and many staff were now waiting to see what technology they would be required to adopt next as part of the move towards a national approach to I.T. strategy.

Case B

The second case study, Force B, also covers a city, together with a large area of countryside, which encompasses several towns. The technological infrastructure in 1993 was seen as a significant factor in spurring the development of an I.T. strategy, in that the lease for the mainframe-based system was due to expire in 1995 after some 15 years of usage. The people with influence over ISS in 1993 comprised a small group of police officers (a Chief Superintendent, a Superintendent, and a Chief Inspector) within the I.T. Department and the civilian information systems (I.S.) manager. The executive level were perceived by the senior police officers comprising the I.T. group as being disinterested in I.T.

The common view of those spoken with was that the concept of I.T. strategy only came to be considered by Police Forces in the early 1990s following the publication of a series of Home Office circulars (e.g. Home Office, 1983, 1993) and Audit Commission papers (e.g. Audit Commission, 1993) on strategy and I.T. These factors, together with inspection visits by the Inspectorate, were said to have 'concentrated the mind' of many senior staff.

In 1993, the I.S. Manager suggested that they should consider developing an I.T. strategy for the next few years. As the Superintendent in the Department noted:

we never had a strategy that has been documented and stated in the past, it evolved through a series of separate developments and it is only now that we are trying to pull them together.

In the I.T. department, I.T. development was seen as a response to the ending of the mainframe lease in April 1995. The applications running on the mainframe were written in the COBOL, which the Force could not support, and they were not millennium compliant. Viewed in combination, these factors were said to have highlighted to the senior staff in the I.T. department that they needed to develop a strategy for future hardware and applications. Other senior officers, outside of the I.T. department, seemed unaware of the lease expiry, or were dismissive of it as a stimulant to any I.T. initiatives. A common view expressed was that the existing systems were in a mess, and that it was about time that something was done about them. The following comments from a Chief Superintendent reflect the views expressed by many others

you only have to look at the hotchpotch of different computer systems that we've got, that we've built up over the years to realise that this is how to build up an information gathering system when you hadn't really thought about what you want to

achieve in the first place. I mean, it's a dogs breakfast, and I mean I remember how we got to be the way we are and it was all to do with money being available, money being available quickly ... 'we've got to spend that money, let's get something on board here ... pressures from persuasive personalities within the organisation can result in them having some sort of stand alone computer system which doesn't merge in with the strategy that we're operating on, and yes we've got a dogs breakfast when it comes to information systems.

In 1994 a project board was set up to monitor the I.T. strategy. The senior police officers with operational responsibilities that were spoken with between 1994 and 1998 viewed this as a necessary device to monitor the project given the poor track record in delivering I.T. projects in the past. The Chief Superintendent and the Superintendent in the I.T. department, who had greater responsibility for I.T., viewed the project board as signifying a lack of trust in their ability to deliver. However, all of those spoken with in ranks of Superintendent and above were of the view that such procedures were necessary to satisfy the expectations of external agencies, such as the Inspectorate and the Audit Commission. In 1994, it was generally agreed that for an I.T. development to be accepted, there must be a strong emphasis upon assisting operational Police officers. The Deputy Chief Constable, as the most senior officer on the project board, summed up what he was seeking

you build up your I.T. strategy by saying 'what is the bloody strategy of the Police Force, what actually are we here for', We're here, I think, to lock up bad people, look after good people, and keep order on the street. We do that by getting the maximum resources employed at the sharp end by keeping paperwork and bureaucracy to a minimum, by using civilians at every possible post we have and by denying Police officers every excuse you can find to come in out the rain and that sort of thing. You then develop your I.T. strategy around that philosophy.

At this point, members of the project board envisaged I.T. strategy as a five year plan, although as was noted,

I would think that every year you would want to stop and reassess your ... not change direction but perhaps the course, because perhaps what someone asks for now is not technically feasible but in two years time it may well be, so you've got to have the flexibility in the strategy to deal with that.

By 1995, most people were reassured that the documented strategy had been thoroughly thought through, and that it reflected trends in industry. The executive level officers reassured themselves of this by engaging a nationally known firm of I.T. consultants to provide a view on the work done to date in developing the strategy. Following this, a request for expressions of interest in supplying the required applications was then sent out. Following submissions from 12 companies, a shortlist of three possible suppliers was drawn up by the I.S. manager and several senior police officers from both the I.T. department, and from the project board. Three meetings were then conducted between members of the Project Board and the shortlisted companies during May and

June 1995. One firm was then appointed as the lead supplier with contract negotiations continuing until September 1995 when a contract was signed for the delivery of 10 applications and a new data network. The budget for hardware and software was £3.3 million, with a further £500,000 being allocated for the network. The applications were not bespoke, but did require customisation.

Years 1996 and 1997 were spent working with the suppliers in developing the new applications and preparing for implementation. The attention given to other areas of I.T. were minimal, and the I.S. manager found it increasingly difficult to gain acceptance for the view that I.T. development needed to be monitored on an ongoing basis. In the view of senior police officers, once a contract had been signed with the supplier, it was a question of sitting back and waiting for the system to arrive. During this time, periodic updates were submitted to the Project Board by the I.S. manager, although the meetings were seen as uneventful and unproductive.

In December 1996, the removal of police officers from the reporting structure between the civilian I.S. manager and the executive level (i.e. Assistant Chief Constable upwards) was regarded by all those involved as being beneficial to the process of decision making. Several staff commented that the fact that police officers were feeding information to the executive officers on I.T. matters served to complicate the issues because of the lack of expert knowledge by both parties. Aside from those directly affected by such changes, most other staff spoken with regarded this as likely to be beneficial to future I.T. usage, with the removal of police officers from areas that 'they knew nothing about'.

Between May 1996 and July 1998, five software applications were implemented in the Force as part of the I.T. strategy, these being crime recording, crime reporting, criminal intelligence, personnel, and command and control. In addition to a new wide area network, approximately 700 personal computers were installed. Progress in implementing I.T. applications was perceived to be good, although the envisaged timescale had slipped due to the diversion of resources into planning for some major operational activities in the Force area.

Case C

Within the third case study, Force C there had been a history of under-investment in I.T. throughout the 1980's and early 1990s. Many staff referred to 'the I.T. disaster', when a large mainframe-based system was procured in the late 1970s, and which retained a 'folklore' status in the late 1990s. As one Chief Superintendent commented

it was an absolute disaster and did not work, so people got their fingers burned ... and the then Inspector was getting most of the blame. Whether it was the system that was intrinsically wrong, whether he should have spotted it, whether it was a lack of ability of foresight I've no idea I'm just glad it wasn't me.

A new Chief Constable arrived 1990, someone viewed by other senior officers as having a strong interest in I.T. At the time of the new Chief Constable's arrival, the I.T. situation was described by several people as being totally uncoordinated in terms of development, purchasing, maintenance, and funding. In 1990, the Chief Constable ordered a review of I.S. in the Force, with a view to developing an ISS for the Force. This review was undertaken by a mid-ranking police officers responsible for I.T., and delivered 1 month later. The conclusion of the review was that the Force should start again with I.T.

In February 1991 the Chief Constable asked the Officer who had undertaken the review to develop a strategy for a new I.T. infrastructure for the Force. The strategy was drawn up over the next few months in an approach that was described as 'high level strategy, very conceptual'. Many senior Officers in the Force commented upon this being the first time that I.T. had been considered strategically within the Force. The I.T. strategy document was submitted to the Chief Constable in the form of a two-page briefing report. This document formed the basis of an operational requirement that was submitted to a local supplier with whom the Force now had a relationship. This relationship arose from an association that the new Chief Constable had with the company in a previous role. In 1992, the supplier began working to develop a corporate database system that was networked throughout the Force. The overall business requirement was not documented within the Force, but rather it was left to the supplier to maintain records of the requirements as discussions with representatives of the Force progressed. The relationship was regarded as developmental, with the supplier developing the applications based upon existing corporate database software, with a view to selling the applications to other Forces. The general process informing the I.T. developments was described by a Police Inspector who was part of the I.T. team as follows:

The general thought process in this Force when we started was that ... well, the general opening statement to the business case to senior management was that we're going to present this thing to you, we'll outline the concepts and the general business case of how we see it working out. We will take on board your observations, but unless those observations are reflected in the vast majority of the workforce, your observations will be discarded, because in five years time none of you, or very few of you will be here working with the system, ... it will be middle management, junior management that will be occupying your position in five years time, and what we don't want to happen is that in five years time somebody says, 'who the fuck put this in?'... 'oh well it was so and so'... 'where is he?'... 'oh he's retired'... No!

The system went live across the Force in 1993, so that by the end of 1993, all operational police officers in the Force had access to a system terminal. Despite the availability, there was considerable unhappiness among many senior staff regarding the performance of the system, and the changes in practice that ensued as a

result of police officers having to use the system. One Superintendent observed

I think the Chief Constable was pushing it that much that there was a general fear of anyone daring to criticise [the] wonderful [system] ... the all being all speaking machines. I think there was a reluctance initially, but of course you hope that these Officers, who are Police Officers who are developing the system, would have got it right in the first place. But unfortunately they didn't.

Between 1995 and 1997, the system was subject to further development to improve functionality and features. These developments, together with ongoing maintenance, continued until 1998, when the I.T. budget was withheld to allow for spending on other commitments in the Force.

Throughout the period from 1994 to 1998, numerous senior Officers expressed their unhappiness about the lack of a documented I.T. strategy, however, they felt unable to do anything about it. They were particularly concerned by the prospect of the Chief Constable leaving to take up another post, which in their view would mean that the strategy would be in jeopardy should the next Chief Constable not approve of it. While they were pleased to have some technology in place that appeared to work, they were of the view that the I.S./I.T. Project Board (as it was termed in this Force – but generally referred to as the 'is it' Board), set up in 1996, should be more proactive in developing a documented strategy, although they saw little sign of that happening. By this they meant that there was nothing written down that expressed what it was that was supposed to be being achieved with I.T., where I.T. developments were going, or what the timetable was. The technology was viewed as being heavily dependent upon the views of one senior officer (i.e. the Chief Constable), which did not necessarily reflect the views, or requirements, of other staff in the organisation. Those closely involved in the management of I.T. in the Force, acknowledged that the rapid pace of I.T. development had meant that they had not produced an I.T. strategy as such, but that they preferred to maintain a verbal agreement upon, and, so they felt, a shared understanding of I.T. strategy.

In 1997, the I.S./I.T. Board met once a quarter, but was perceived by the senior police officers taking part to have become redundant, as they knew that there was no money available to develop I.T... I.T. strategy in the Force remained undocumented. In mid-1997 an inspection by the Inspectorate of Constabulary was highly critical of several facets of the I.T. operation, namely, and in particular the lack of a documented I.T. strategy. In addition, the auditors expressed surprise that there was no business plan for the Force.

Following this, I.S./I.T. Project Board, and the Chief Constable, decided that they must be seen to be addressing the criticisms made by the Inspectorate body in a recent report. In late-1997 the Chief Constable authorised two senior officers to draw up a business plan for the Force, with the I.S./I.T. Project Board, and the I.T.

department being asked to submit its intentions into this process. In April 1998, the I.S./I.T. Project Board, in consultation with the I.T. department and the Chief Constable, decided that no further development should be undertaken of the system, other than that required for maintenance. The reason for this was said to be the ongoing development of a national I.T. strategy, and it was decided that they should wait to see what emerged from the national strategy with a view to aligning their systems with it. By late-1998, the I.T. strategy had still not been documented, and it was at this point that the I.S./I.T. Project Board was disbanded following the retirement of the Chief Superintendent who chaired the Board.

Worlds and objects in the shaping of I.T. trajectories

We consider the trajectories of I.T. and change in the three cases, highlighting the ways in which the shaping of practice was subject to a wide range of interactions, within and across social worlds. In doing this we have chosen to focus our discussion upon boundary objects and social worlds, because of the ways in which these became influential in understanding what took place. In each of these case studies, new information technologies (hardware and software) were introduced – that is, an array of artefacts embedded within social and work practices that was unique to each case situation. In cases A and B, the software procured consisted largely of pre-existing applications that required customisation, in the process becoming complex configurational technologies (Williams, 1997) that to varying degrees reflected the specific context. The same could be said of case C, where the applications were developed by the suppliers as applications to be sold off-the-shelf to other, similar, organisations. However, the point of concentrating upon the notion of trajectory is that it focuses attention on the I.T. initiatives as continuations of practices surrounding I.T. within each of the cases. The idea of trajectory encourages consideration of the antecedents of the initiative in focus, as well as practice associated with the more immediate activities, interactions and events.

The antecedents were important, as it became evident that in each of the cases the apparent perceptions of I.T. initiatives from the 1970's and 1980's differed among many senior staff, for whom these initiatives still provided a common reference point when talking about I.T. As one Chief Inspector discussing earlier I.T. initiatives in case A commented:

it was an absolute disaster and did not work ... I'm just glad it wasn't me.

These I.T. 'disasters' acted as abstract boundary objects, with espoused meanings partially shared and partially divergent among groups. The divergence was evidenced through additional meanings that different groupings discussed. For example, in all three cases, operational (uniform patrol and detective) officers tended to discuss the earlier I.T. initiatives, or 'disasters' to them, as

representing what happens when non-operational police officers are put in charge of such areas. As a Chief Superintendent, this time in case B, reflected

I mean I remember how we got to be the way we are and it was all to do with money being available, money being available quickly ... 'we've got to spend that money, let's get something on board here' ... pressures from persuasive personalities within the organisation can result in them having some sort of stand alone computer system which doesn't merge in with the strategy that we're operating on.

Alternatively, non-operational police officers who were involved with I.T. in each case tended to view the earlier 'disasters' as reflecting what happens when there was no clear strategy, implicitly justifying their own contributions to the ongoing I.T. initiatives. This difference highlighted both the differences in usage, but also a distinction between social worlds – the operational and the non-operational police officers; with the former, almost universally, speaking disparagingly of the latter. Several senior police participants were wary of becoming too closely involved, or aligned with the I.T. projects when 'history', as they saw it, had shown it to be disadvantageous to career progression to be associated with an 'I.T. disaster'. Senior police officers directly involved with the I.T. projects also expressed concern with how they were perceived, and the implications for career progression, except where they were nearing retirement. For the majority of those spoken with, those engaged within the social world related to the I.T. initiatives either represented civilian 'experts', or non-operational police officers who were '*meddling in things they don't know about*', as an Assistant Chief Constable put it.

These social and symbolic facets of practice are important for our understanding of the outcomes, having implications for the material objects of the I.T. initiatives. They represent a part of the way in which technology and social worlds go through processes of mutual adaptation (Orlikowski, 1992); where I.T. initiatives deriving from activities apparently controlled by, in these cases, civilian 'experts' seemed to be accorded the label of 'likely to be useful to me as the operational officer' (in the latter stages of case B following the appointment of a senior, civilian, I.T. manager). In contrast, I.T. initiatives emerging through a process apparently controlled by police officers, no matter how senior, were labelled as likely to be of 'dubious use' (cases A and C), before any new I.T. hardware or software had appeared. Thus, it seems here that perceptions of history, specifically the boundary object of 'the disaster', encouraged or discouraged enthusiasm for participation in the social worlds associated with the I.T. initiatives. Furthermore, perceptions of the social worlds tasked with planning and procuring the I.T. as 'expert' or otherwise were influential in shaping views related to the likely usefulness of the outcome of each I.T. initiative. This suggests that when looking at the detail of organisational activity, viewing

I.T. trajectories simply as a series of discrete stages, or phases, would have been inadequate. The concept of boundary objects has drawn attention to the way in which these constructs 'linger' within and across the activity of social worlds for extended periods of time, being actively referenced by people, and apparently influencing areas as diverse as participation and non-participation, as well as interpretations of usefulness of I.T. artefacts.

Façades of formality?

In each of the cases, 'the I.T. strategy' was a term used frequently by almost all those spoken with, and the ideas captured by such a term (terms reflecting an area of perceived activity for many – because in cases A and C there was no 'physical' I.T. strategy that could be referred to) could also be seen to act as abstract boundary objects in each case. These boundary objects appeared to provide the impetus for the sustaining of the temporary social worlds that constituted the people involved in delivering the I.T. initiative. The practices in each of the cases differed in terms of the representations of formality constructed within these social worlds. According to an Inspector on the I.T. team in case A

it's all in our heads, we don't have time to sit down and write a strategy.

In case C, a Chief Inspector who was a lead member of the group leading the I.T. initiatives discussed the lack of any documentation, saying

I mean the strategy itself, we know has to be not so much part of an iterative strategy but a completely flexible strategy. Part of the problem of strategies is it takes so long to get the bloody things in place that to try and adhere to it two years down the line you're cutting your arm off because you say 'I cannae use that 'cause it's not in the strategy and we've agreed that the strategy cannot be altered'. Not quite.

In contrast to this, in case B, developed considerable I.T. strategy documentation through a lengthy series of formal planning, and project management meetings – with these formalised practices constituting readily identifiable trajectory schemas. However, it was not that these more formal trajectory schemas were lacking in case A, but rather that they each had different forms of trajectory schema in place. It has been frequently observed that I.T. initiatives often comprise both formal and informal practice (Sillince & Mouakket, 1997). Staff in cases B and C were quite open that the use of formal planning meetings and project management methods was a means of ensuring favourable comment when audited/inspected. Similarly, staff in case A bemoaned the fact that they were repeatedly criticised for the lack of such formality, irrespective, in their view, of the success of their informal approach to appropriating and implementing I.T. hardware and applications. Staff in case A adopted more formal practices, in 1997, but as one Chief Superintendent commented

we were supposed to be looking at following PRINCE, but it is very much in name only, we're certainly not doing all the documentation. ... We don't have any methodology really, we just have to get on with it.

In other words, despite not believing in the need to formalise what they were doing, nonetheless, the potential for adverse comment when next inspected or audited meant that effort was put in to creating an impression of a formal process surrounding the I.T. developments. It is feasible to consider such attempts to incorporate formality something of a façade, masking the more informal practices constituting the activity of the groups working on the I.T. initiatives. This does not mean that such formal façades should be dismissed – quite the contrary; they were part of the shaping of the I.T. trajectory. The point is that the people in two of the case studies went out of their way to construct these 'formal' processes. It was the existence, and likely future work of the public sector audit and inspection bodies, as well as the results of previous inspections, acting as both abstract (the former) and physical (the latter) boundary objects that stimulated these processes. The frequent reference to these boundary objects intimated their role in shaping practice related to the I.T. initiatives.

A similar phenomenon has been noted elsewhere in the U.K. public sector, where a lot of managerial time and effort has gone into formalising an abstract rationality of objectivity (Hanlon *et al.*, 2005) in response to selected boundary objects. Those working in each of the case organisations may have wanted the I.T. outcome to be successfully incorporated, or to achieve configurational 'fit' (Williams, 1997), but they wanted both the outcome *and* the process to receive the stamp of approval when they were next audited/inspected. However, we do not see the practices in the cases reported here as façades; rather, they constitute a part of the trajectories. We would not argue that as areas of practice they were any less influential in the ongoing interactions of social worlds in shaping socio-technical outcomes.

Conclusion

The ways in which people in organisations go about deciding upon, acquiring and introducing large-scale I.T. initiatives is predominantly a matter of social interaction, which influences and is influenced by meanings and interpretations. While the contextually specific setting of the U.K. Police Service makes for a fairly unique sphere of operation, the findings may well have similarities with other areas of activity, particularly in public sector environments.

In these three cases we have seen considerable variation in both process and outcome. The trajectories of each case differed through practices, which we argue, reflect social shaping of I.T. trajectories. Nonetheless, there are some notable distinguishing aspects between the cases. Two of the cases (cases A and C) ended up with technology that was perceived, by most of those spoken

with, to have been imposed. The social worlds were seen as being controlled by non-experts, and the views of operational police officers in related social worlds were largely negative regarding the likely usefulness of the I.T. The outcome was that in each case the technology was not well received, and replacement applications were sought after a fairly short space of time (within 1 year). By contrast, in case B the technology that ensued was widely anticipated, and was generally well received. The social world here was viewed as being controlled by 'experts' (senior civilian staff).

Our findings in these cases do not support discussions of I.T. choice/decision-making/strategy that focus upon bi-polar distinctions between planning and serendipity. In each of the case studies reported in this paper, planning was evident, but then we note that effort was put into formalising a rationality of objectivity (Hanlon *et al.*, 2005), ostensibly for the benefit of external audit and inspection bodies. This would seem at first glance to

support a link between institutional type and approaches to choosing and implementing I.T. However, as we have argued, much of the formal planning was, to a large extent, solely for the benefit of external audit and inspection bodies, which could be viewed as boundary objects within and across the social worlds in each case. We have noted already that a similar phenomenon has been observed in other areas of the U.K. public sector activity. How widespread this may be, and whether it should be of concern are areas for future work. These are not trivial questions, especially in view of the efforts being put into the U.K. Government's Gateway Process for I.T. projects, and the previously mentioned concern with RFS among public sector I.T. initiatives. The additional significance of this study is to argue that there is potential in considering boundary objects and their interpretations within and across social worlds, as in these instances, these 'objects' were significant in the shaping of I.T. trajectories.

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